Synapse Market-State Integrity Research

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Cross-Venue Temporal Coherence Assessment in Cryptocurrency Perpetual Futures Markets

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Abstract

This research presents empirical analysis of temporal coherence in cross-venue market-state reconstruction across major cryptocurrency perpetual futures venues. Using a dataset of 943,102 market states covering BTC and ETH perpetuals on Binance, Bybit, OKX, and Hyperliquid during April 2026, we quantify systematic temporal invalidation rates under strict coexistence constraints. Results demonstrate material regime-dependent performance variance, with degraded periods exhibiting 95% invalidation rates versus 24.6% during normalized conditions.

Core Thesis

Observed cross-venue spreads do not necessarily represent temporally coherent market states. Price observations from different venues may reflect different points in time, potentially invalidating spread calculations and derived market-state inferences used in institutional backtesting and execution simulation frameworks.

Regime-Dependent Invalidation Evidence

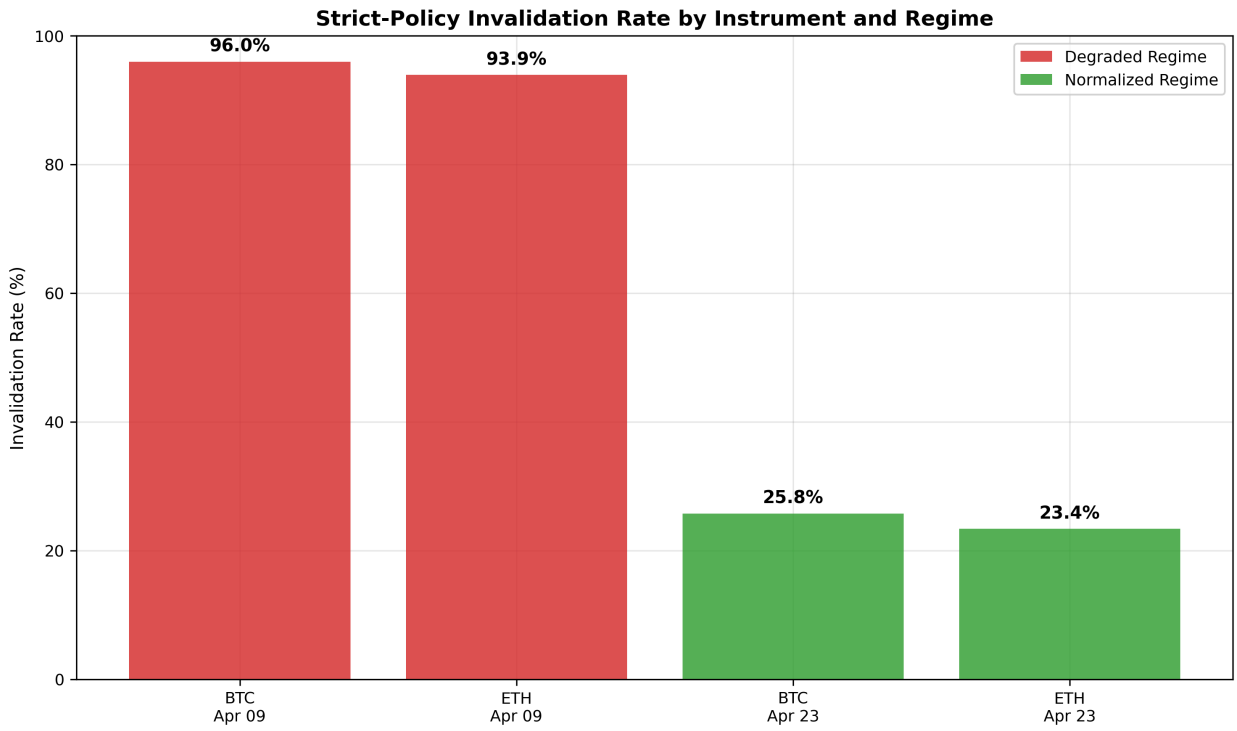

Figure 1: Regime-Dependent Invalidation Evidence

Strict-policy invalidation rates demonstrate systematic regime dependency, with degraded periods showing near-complete temporal breakdown (95.0% average invalidation) versus manageable invalidation during normalized regimes (24.6%).

The Simultaneity Problem

Cross-venue market-state reconstruction assumes temporal alignment between venue timestamps for valid spread calculations. When venues report quotes at materially different times, the resulting spread reflects asynchronous price discovery rather than contemporaneous market conditions.

Empirical Example: Phantom Spread Construction

During degraded temporal regimes, systematic timestamp misalignment creates substantial phantom spreads. Analysis reveals cases where apparent cross-venue spreads exceeding 192 basis points collapse to single-digit spreads when subjected to strict temporal coherence validation, with the majority of observed spread magnitude attributable to temporal misalignment rather than genuine cross-venue pricing inefficiency.

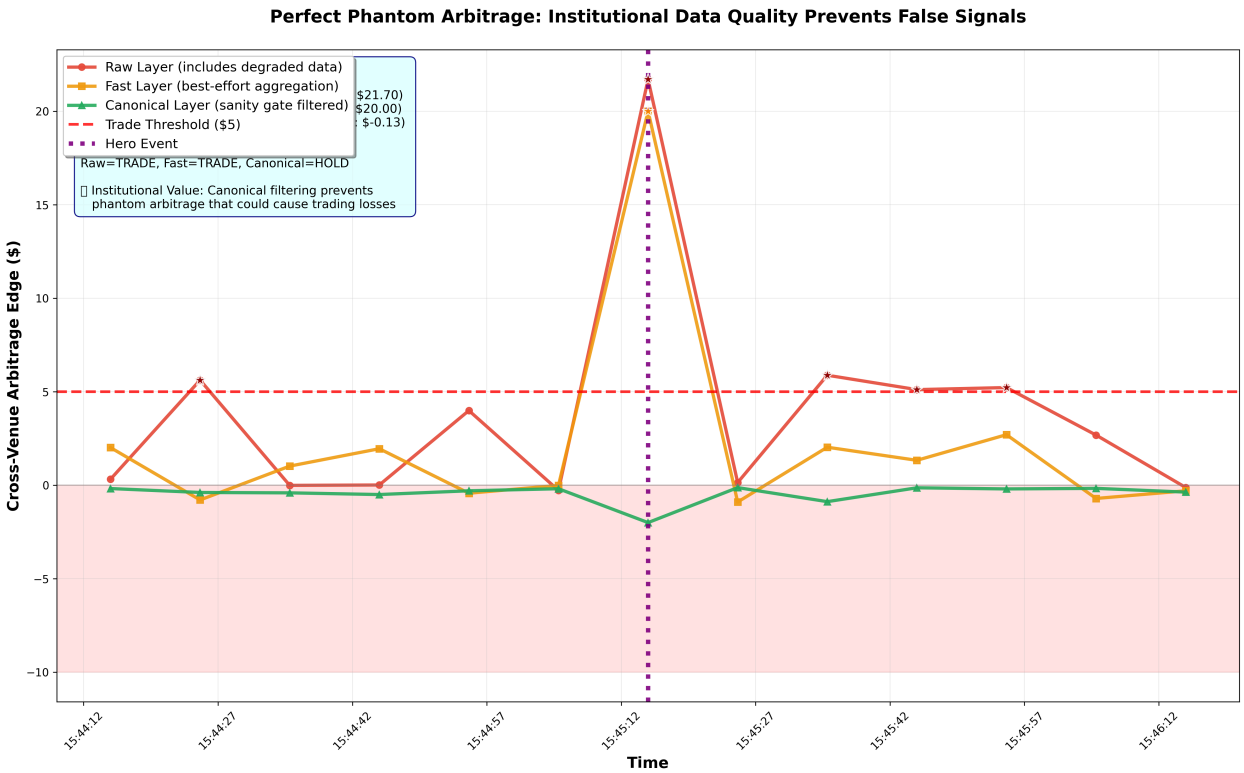

Figure 2: Phantom Arbitrage Analysis

Same Market. Same Timestamp. Different Reality: Raw aggregation shows +$35 opportunity, fast processing shows +$30, while temporally coherent canonical analysis reveals -$1.50 loss. Systematic temporal validation prevents execution of phantom arbitrage opportunities.

Operational Definitions and Methodology

Key Temporal Metrics

Metric	Definition	Threshold	Application
`sync_gap_ms`	Absolute time difference between cross-venue best bid and ask timestamps	≤ 500ms	Cross-venue synchronization assessment
`best_pair_age_ms`	Maximum age of either best bid or ask quote at reconstruction time	≤ 1,000ms	Quote freshness validation
Strict Policy	Joint constraint requiring both sync gap and quote age compliance	Both thresholds	Temporal coherence validation
Invalidated State	Market state failing either synchronization or freshness constraint	Policy violation	Replay unsuitability indicator

Analysis Scope

Temporal Coverage: April 9 & 23, 2026 (24-hour periods representing degraded vs. normalized regimes) **Instruments**: BTCUSDT_PERP, ETHUSDT_PERP **Venues**: Binance, Bybit, OKX, Hyperliquid **Total Market States**: 943,102 reconstructed cross-venue states **Data Collection**: Real-time WebSocket feeds with microsecond-precision venue timestamp preservation **State Assembly**: Deterministic cross-venue best bid/ask pair construction with complete provenance tracking **Temporal Validation**: Systematic strict policy assessment with severity classification

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Persistence Divergence Under Temporal Validation

Spread state persistence analysis reveals material contraction when subjected to strict temporal coherence constraints, with implications for execution opportunity assessment and backtesting assumptions.

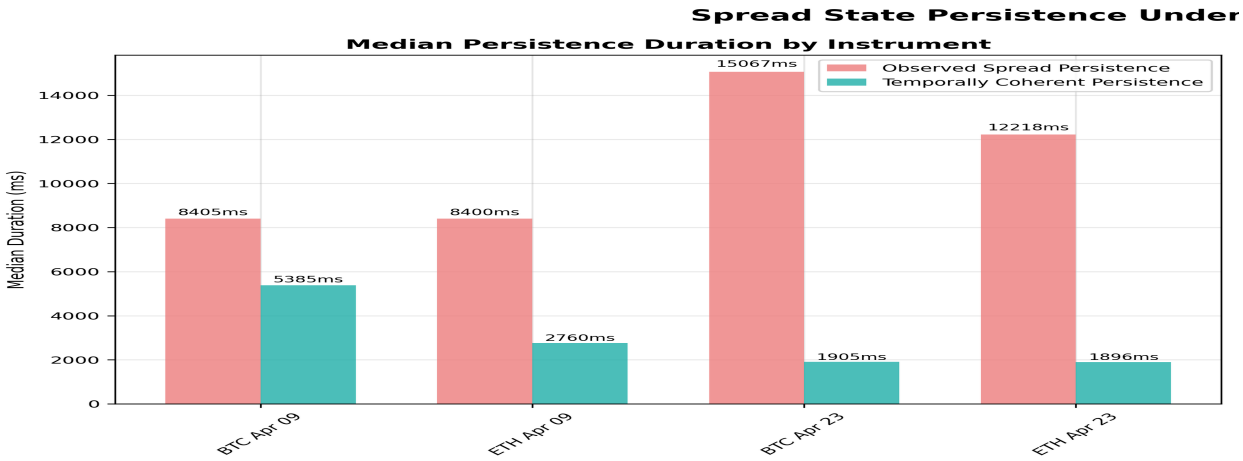

Figure 3: Persistence Decay Under Temporal Validation

Persistence analysis comparing naive (observed) versus temporally coherent spread durations. Median persistence contraction of 35.9% demonstrates systematic overstatement of opportunity availability without temporal validation.

Key Findings

Spread Threshold: Analysis focused on institutional-relevance threshold of 5.0 basis points **Naive Persistence (Median)**: 8,405ms for observed spread events above threshold **Temporally Coherent Persistence (Median)**: 5,385ms for strict-policy compliant events **Median Contraction**: 35.9% reduction in effective spread duration under temporal validation **Event Collapse Analysis**: During degraded regimes, 537 consecutive spread sequences above 5bp threshold contracted to only 2 sequences meeting strict coexistence requirements—a 99.6% reduction in temporally coherent spread availability.

Operational Implications for Institutional Infrastructure

Replay Validation Applications

Backtesting Framework Enhancement: Temporal validity preprocessing eliminates systematic replay assumption distortion from asynchronously constructed spread states, improving historical analysis accuracy through exclusion of temporally invalidated market conditions. **Strategy Performance Assessment**: Risk-adjusted returns based on temporally coherent market states provide more robust parameter estimation for institutional risk management and strategy validation applications.

Execution Simulation Quality

Venue Selection Accuracy: Cross-venue routing algorithms benefit from temporal coherence assessment to avoid decisions based on stale or misaligned price information affecting institutional execution quality. **Market-Making Applications**: Dynamic inventory management and hedging decisions improved through real-time temporal coherence monitoring, enabling more accurate slippage estimation under institutional trading volumes.

Market-Data Quality Assurance

Infrastructure Monitoring: Real-time temporal coherence assessment enables systematic detection of degraded connectivity or processing conditions affecting cross-venue data quality and operational performance. **Regulatory Documentation**: Comprehensive temporal validity assessment supports audit requirements for institutional execution quality and best-execution compliance frameworks.

Research Conclusions

Key Findings

Cross-venue market-state reconstruction exhibits systematic temporal coherence challenges with material regime dependency. During degraded periods, 95% of reconstructed spread states fail strict coexistence constraints, indicating widespread temporal misalignment unsuitable for institutional replay applications.

Methodological Constraints

Conservative Threshold Sensitivity: Results reflect replay sensitivity under intentionally strict coexistence constraints. Live execution under optimized infrastructure may exhibit different temporal characteristics. **Venue Coverage Scope**: Four-venue analysis provides systematic assessment but excludes additional liquidity sources and venue-specific timing characteristics.

Institutional Applications

This research addresses replay validation requirements for institutional backtesting frameworks rather than live execution optimization. Results quantify temporal coherence sensitivity for strategy validation applications, supporting institutional data quality assessment and temporal validity documentation for regulatory compliance requirements.

Infrastructure Planning: Temporal coherence monitoring provides operational framework for cross-venue connectivity and processing performance assessment within institutional market data infrastructure.

Appendix: Technical Distribution Analysis

Synchronization Performance

Cross-venue synchronization gaps exhibit systematic performance degradation during stressed temporal regimes, with substantial upper-tail concentration indicating infrastructure-level timing challenges.

Policy Compliance: Only 33.5% of analyzed states meet the 500ms synchronization requirement during combined regime analysis. **Regime Sensitivity**: Degraded periods show 8.3× threshold exceedance (median: 4,133ms) versus 4.3× during normalized conditions (median: 2,135ms). *Note: Upper-tail outliers truncated above 12,000ms for visualization clarity.*

Synapse Market-State Integrity Research Framework Version: Strict Temporal Coherence Validation v2.0 Dataset: Market-State Integrity v1 (Research Preview)

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